

Equity in Transition: Analyzing the Distributional Impacts of Low-Carbon Energy Technology Subsidies

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1 Introduction

One concern about low-carbon (“clean”) technology subsidies is that high-income households claim them more due to high initial costs. For example, on average, the top income quintile in the United States (US) received more than 50% of investment tax credits for residential energy efficiency improvements since introducing these credits, according to the Internal Revenue Service’s Statistics of Income data. If these subsidies that mainly go to richer households are funded by general taxpayer revenues, then they are pecuniary transfers from poorer to richer households. Such transfers are perceived as regressive in monetary terms and raise equity concerns among the public.

These concerns are particularly salient under the Inflation Reduction Act (IRA) of 2022, which extended and expanded the Investment Tax Credit (ITC) for residential clean energy installations—such as rooftop solar systems—by offering a 30% nonrefundable credit for eligible expenditures through 2032. While the IRA represents an unprecedented federal commitment to decarbonization, its current structure raises important questions about who benefits most from clean energy subsidies and how these subsidies affect economic inequality. Most low- and middle-income households, for example, may be unable to claim these tax credits due to limited tax liability, making the benefits of the policy highly skewed toward wealthier households.

At the same time, if we consider the environmental co-benefits, learning-by-doing externalities, and general equilibrium effects, these subsidies—although concentrated among the wealthy—may generate broader welfare benefits that disproportionately help lower-income households. For instance, reducing local air pollution associated with decreased greenhouse gas (GHG) emissions can yield larger health improvements for poorer households who are more frequently exposed to pollution. Moreover, since high-income households are responsible for a disproportionate share of residential emissions, accelerating their adoption of clean technologies can amplify environmental gains. Finally, subsidizing early adopters—who are often wealthier—can reduce the technology’s costs through learning-by-doing, eventually making adoption feasible for lower-income households.

Motivated by this apparent regressivity concern and its relevance under the IRA, I ask: How is low-carbon technology adoption related to household income in the US? Focusing on residential rooftop solar panel adoption, I document a positive correlation between income and adoption rates. If the richer households disproportionately receive these subsidies, then what are the distributional welfare implications of clean technology subsidies funded by general taxation? To answer this, I calibrate a model to reflect the early stages of the energy transition in the US economy and use it to quantify the equity and efficiency implications of IRA-era residential clean energy subsidies.

Finally, I ask what the best policy mix is for achieving multiple objectives—emissions reduction, welfare maximization, and speed of transition—while maintaining majority political support. I use the calibrated model to evaluate a set of alternative policy designs and financing mechanisms, including a uniform labor income tax, a progressive income tax, a corporate income tax, an emissions tax, and government debt. I also consider alternative subsidy program structures, such as income-capped or refundable tax credits, to assess their effectiveness and equity. These experiments aim to inform the implementation and refinement of IRA and post-IRA clean energy subsidy programs in the US and abroad.

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2 Literature Review

Versions of the questions posed here have been addressed in literature. Most of the previous research falls into two categories: (i) empirical analyses of the distributional effects of environmental policies, and (ii) the macroeconomic studies on the unequal economic consequences of climate change policies.

Borenstein and Davis (2024) have well documented the adoption patterns for various low-carbon energy technologies and tax credit receipts for these technologies across income groups in the US. However, their study neither goes beyond descriptive facts nor quantifies dynamic general equilibrium effects, such as the indirect effects on relative prices, or the nonpecuniary effects such as the health co-benefits of reducing local air pollutants. Vona (2023) summarizes the multidimensional impacts that climate policies have on well-being which have been documented by various researchers and provides an overview of the empirical literature. The consensus of this empirical literature is that subsidies to clean and energy-efficient equipment exacerbate the regressivity of climate policies in the presence of financial constraints by benefiting the wealthier households at the expense of less wealthy ones.

I will contribute to these empirical analyses of who benefits from clean technology subsidies by quantifying the overall distributional effects of these policies while taking into account multi-dimensional effects using a model calibrated to reflect the US economy. Specifically, I incorporate two additional indirect distributional effects of clean technology subsidies into a general equilibrium model: (i) the health co-benefits of reducing local air pollutants associated with reducing greenhouse gas (GHG) emissions, and (ii) the learning-by-doing spillover effects of early adopters. Reducing GHG emissions has well-known positive effects in terms of health co-benefits such as reducing local air pollutants and improving health outcomes. Banzhaf et al. (2019) summarizes the large body of economic literature that documents a strong positive correlation between ambient air pollution, poverty, and race. The second dimension that I aim to incorporate in my analysis is the learning-by-doing spillover effects from early adopters, who are likely to be wealthier. Gao et al. (2022) is one of few empirical economic analyses which finds that learning through economies of scale in production leads to cost reductions, not only in hardware costs, but also in non-hardware costs in US residential solar installations.

There are several quantitative macroeconomic studies that analyze the distributional effects of climate policies in a general equilibrium framework. The majority of these studies focus on the distributional effects of carbon pricing instead of low-carbon technology subsidies like my paper does. Känzig (2023), Benmir and Roman (2022), Fried et al. (2024), Fried et al. (2018), and Belfiori et al. (2024) are some recent studies that analyze the distributional effects of carbon pricing.

Kuhn and Schlattmann (2024) is the closest work to this project in that they develop a quantitative life-cycle model with heterogeneous adoption rates of carbon-neutral commitment goods by income to quantify the reduction-redistribution trade-off of different policy mixes. Lanteri and Rampini (2025) is another similar macroeconomic study of adoption of clean technologies, but instead by heterogeneous firms, the study uses a dynamic general equilibrium model of firm dynamics and clean technology adoption with financial constraints. Through my review, I have yet to find a quantitative macroeconomic study that includes the health co-benefits and learning-by-doing spillover effects in the distributional analyses of climate policies for heterogeneous households that I propose here.

3 Methodology

I answer the three questions posed in the introduction using a combination of empirical and quantitative methods. Each step builds on the previous to deliver a comprehensive analysis of the distributional impacts of clean technology subsidies under the Inflation Reduction Act (IRA).

First, I compile and summarize stylized facts about the relationship between household income and key clean energy outcomes in the United States. Using cross-sectional data, I document empirical patterns in (i) the share of aggregate energy consumption, (ii) the share of aggregate emissions from residential energy consumption, (iii) residential rooftop solar panel system deployment rates, (iv) the receipts of residential energy credits, (v) ambient air pollution exposure, and (vi) the share of energy expenditure in total expenditure. I provide descriptive summary statistics across the 2015 income quintiles in the US for items (i)-(vi) in Table 1 with descriptions of the data sources used. The data shows that income is positively correlated with residential rooftop solar panel system deployment rates and negatively correlated with mortality damages incurred per capita due to ambient PM2.5 air pollution. I use these data to calibrate the household heterogeneity in the structural model.

The financial benefits of deploying residential rooftop solar panels are substantial for households, as evidenced by regression analyses using data from the 2020 Residential Energy Consumption Survey (RECS) conducted by the US Energy Information Administration. These analyses estimate the impact of on-site solar power generation on household-level annual electricity and total energy expenditures. The results, based on weighted least-squares regressions with clustered standard errors, indicate that households with solar panels spend over \$700 less per year on electricity compared to those without, representing a 47% reduction relative to the average annual electricity bill of approximately \$1,500. The reduction in total annual energy expenditures, while smaller in magnitude, remains both statistically and economically significant: households with solar panels spend about \$600 less per year, or roughly 30% less than the average energy bill of \$2,171. These findings underscore the sizable private cost savings associated with residential solar adoption.

The cost of installing solar panels for on-site power generation is a significant barrier to adoption for many households. The cost of installing solar panels includes the cost of the solar panels, the cost of the inverter, the cost of the mounting hardware, the cost of the wiring, the cost of the installation labor, and the cost of the permitting and inspection. The price of solar panel installations has been decreasing over time due to technological advancements and economies of scale, even before accounting for government incentives. While the average cost of installing a median size solar panel system in the US was around 70% annual median gross income in 2010, it has decreased to around 20% of annual median gross income in 2020.

Motivated by these empirical observations, I develop a heterogeneous agent dynamic stochastic general equilibrium (DSGE) model with incomplete markets, building on Aiyagari (1994), with the inclusion of costly clean energy adoption decisions and environmental externalities.

In the model consumers supply an exogenous amount of labor, receive wage income, accumulate assets in physical capital, and rent out capital to firms. Consumption requires energy use, and energy use is assumed to be an affine function of final good consumption. This assumption implies that the expenditure share of energy is larger for households with lower consumption levels, which is an empirical fact documented in Table 1.

When using the old, dirty energy technology energy consumption creates ambient air pollution – a flow variable that results in production damages – a negative externality to society. Consumers can switch to a clean energy source by deploying rooftop solar panels to generate electricity with a one-time investment cost and once they make the switch they thenceforth permanently use the clean energy source. The unit cost of clean energy is lower than the unit cost of dirty energy,

but the initial investment cost is very high. As the scale of clean energy adoption increases, the investment cost decreases due to learning-by-doing spillover effects.

The labor supply is stochastic and is governed by an idiosyncratic shock. Consumers can self-insure by accumulating assets subject to a borrowing constraint. Moreover, households face an exogenous labor income tax rate. The government uses the tax revenue to finance the subsidy for the clean technology investment cost, and returns the remaining tax revenue to households equally in a lump-sum manner. Firms combine capital and labor to produce the final good (used for consumption and investment). The negative externality generated by dirty energy consumption is not internalized by the firms or consumers, but leads to a fraction of output being lost due to damages. The loss of utility will be greater for poorer households due to decreasing marginal utility of consumption. Both firms and consumers act competitively and take prices as given.

The novel component of this model is that it incorporates a discrete adoption choice. A household's consumption, saving, and adoption decision will be governed by three household state variables: the household's risk-free asset holdings a , the household's stochastic labor productivity ℓ , and the household's adoption status $s \in \{0, 1\}$, in addition to one aggregate state variable: the distribution of households across the wealth, income and adoption states $\Phi(a, \ell, s)$. The ambient air quality index is not a state variable because it is assumed to be a flow variable and a function of only the current period's aggregate emissions from dirty energy consumption, and thus, is a function of $\Phi(a, \ell, s)$ alone.

The Bellman equation of a household who has not adopted the clean technology in state $s = 0$ at the beginning of period t is given by:

$$V_t(a, \ell, 0; \Phi) = \max_c \left\{ \begin{aligned} &\max_c U(c) + \beta \mathbb{E}_t \left\{ V_{t+1}[w(1 - \tau_t^\ell)\ell + (1 + r)a + T - c - \bar{q}e(c), \ell', 0; \Phi'] | \ell \right\}, \\ &\max_c U(c) + \beta \mathbb{E}_t \left\{ V_{t+1}[w(1 - \tau_t^\ell)\ell + (1 + r)a + T - c - \bar{q}e(c) - p(1 - \tau), \ell', 1; \Phi'] | \ell \right\}, \end{aligned} \right.$$

subject to $\Phi' = H(\Phi)$,

where c is final good consumption, $e(\cdot)$ is energy consumption function of final good consumption, w and r are wage and interest rate, $\bar{q}$ is the energy price when using the dirty technology, $S \in \{0, 1\}$ is the adoption decision, p is the clean technology investment cost, τ is the exogenous technology subsidy rate, τ^ℓ is the endogenous labor income tax rate levied to finance the subsidy, T is the lump-sum transfer from the government to households, and $H(\cdot)$ is the law of motion governing the distribution of households across the state variables' next period as a function of the distribution in the current period.

In the period that follows the adoption decision, the household will be in the state $s = 1$ at the beginning of period t and have the Bellman equation given by:

$$V_t(a, \ell, 1; \Phi) = \max_c U(c) + \beta \mathbb{E}_t \left\{ V_{t+1}[w(1 - \tau_t^\ell)\ell + (1 + r)a + T - c - \underline{q}e(c), \ell', 1; \Phi'] | \ell \right\},$$

subject to $\Phi' = H(\Phi)$,

where $\underline{q}$ is the energy price when using the clean technology. Importantly, $\bar{q} > \underline{q}$, thus households have an incentive to adopt the clean technology in the long-run.

I externally calibrate the model to match the energy expenditure shares across the income distribution, the unit variable cost of electricity with and without on-site solar generation, and the historic solar investment tax credit schedule and extensions provided under the Inflation Reduction Act. I used the 2010 Panel Study of Income Dynamics (PSID) data to estimate the joint income

and wealth distribution of households in the US and evaluate the model’s performance in matching the joint income and wealth distribution. These results are summarized in Table 2 in the Appendix.

I first developed a preliminary version of the model that excludes environmental damages and learning-by-doing spillovers in order to isolate the direct welfare impacts of clean technology access and subsidies. Using this version, I quantified the welfare effects of introducing clean technology as an option, as well as the effects of offering subsidies for its adoption. The results show that while all households experience welfare gains from the availability of clean energy technology, the benefits of the subsidy policy are concentrated among middle-income and middle-wealth households. Importantly, while support for the introduction of the clean technology itself is unanimous across the population, only 11.2% of households support subsidizing its adoption when the subsidy is financed through general taxation. These findings suggest that while clean technology adoption has broad appeal, the distributional consequences of its financing can substantially affect political feasibility.

In the next steps of the project, I will extend the model to include learning-by-doing effects and reevaluate these welfare effects. I will conduct counterfactual policy experiments to evaluate the performances of alternative policies under various objectives and political feasibility to answer the final question that I posed using the model I developed.

Further extensions of the model will be studied as separate sensitivity analyses. These extensions will include the following: (i) separate energy markets and endogenizing the energy prices, (ii) nonrefundable tax credits, (iii) clean and dirty energy production job types to evaluate job displacement because of the energy transition, and (iv) permanent types with different discount factors to reflect racial wealth gap.

4 Policy Implications

The proposed project has several important policy implications. First, it highlights the need for policymakers to consider the distributional effects of clean technology subsidies. The preliminary results suggest that these subsidies may be regressive, benefiting wealthier households more than poorer ones. This finding underscores the importance of designing subsidy programs that are equitable and do not disproportionately benefit higher-income households.

Second, the project emphasizes the significance of accounting for environmental co-benefits and learning-by-doing effects when evaluating the overall welfare impacts of clean technology subsidies. By incorporating these additional effects, policymakers can obtain a more comprehensive understanding of the true benefits and costs of such subsidies. This approach can help justify the implementation of subsidies that may initially appear regressive but ultimately provide greater overall welfare benefits.

Third, the project provides insights into the optimal financing strategies for clean technology subsidies. The analysis of different financing mechanisms, such as progressive income taxes, emissions taxes, and government debt, can inform policymakers on the most effective and politically feasible ways to fund these subsidies. This information is crucial for designing policies that achieve desired environmental and welfare outcomes while maintaining public support.

Fourth, the project offers a framework for evaluating alternative policy mixes to achieve various objectives, such as emissions reduction, welfare maximization, and transition speed. By simulating different policy scenarios, the project can identify the most effective combinations of subsidies and financing strategies to meet these objectives. A realistic evaluation of existing US policies will incorporate that the residential clean energy subsidies in the US are nonrefundable tax credits, thus making approximately two-thirds of US households ineligible from claiming these subsidies.

Alternative policy evaluations can guide policymakers in designing comprehensive and balanced policy packages that address multiple goals simultaneously.

Additional model enhancements will improve policy relevance. Endogenizing energy prices through multiple energy markets allows evaluation of feedback effects from widespread clean energy adoption. Including clean and dirty energy job types enables assessment of job displacement and the need for complementary labor policies. Incorporating permanent household types with heterogeneous discount factors captures structural racial wealth disparities, allowing analysis of whether race-neutral subsidies exacerbate inequality and how targeted reforms might reduce it.

Finally, the project’s methodology and findings can be extended to other clean energy technologies and contexts. The insights gained from the analysis of residential rooftop solar panel subsidies can be applied to other technologies, such as heat pumps and energy-efficient appliances. Additionally, the modeling framework can be adapted to study the distributional effects of clean technology subsidies in other countries, providing valuable guidance for international policymakers.

Overall, the proposed project aims to provide a thorough and nuanced analysis of the distributional and welfare impacts of clean technology subsidies. By considering a wide range of factors and policy options, the project seeks to inform the design and implementation of equitable and effective clean energy policies that promote a just and sustainable energy transition.

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Appendix

Table 1: Descriptive statistics for the income quintiles in the US

income %ile	bottom 20%	20%-40%	40%-60%	60%-80%	top 20%	top 5%
share of aggregate income	3.17	8.42	14.37	22.83	51.21	22.81
share of residential energy consumption	12.59	18.16	13.07	19.75	24.42	12.01
share of emissions from residential energy	13.43	18.36	13.89	20.20	23.01	11.05
share of rooftop solar adoptors	0.53	3.42	9.26	16.87	43.75	26.19
share of residential clean energy credits	0.48	4.11	4.08	21.75	48.99	20.59
mortality damages per capita (2020 dollars pp)	4,811	3,910	3,103	2,769	2,354	na
share of energy expenditure in total expenditure	8.5	6.4	5.0	4.2	3.3	2.2

Notes: Reported shares and rates are in %ages, except for the mortality damage per capital values, which are in 2017 US dollars. Income quintiles are constructed based on the upper income limits of quintiles summarized in Table B19080 from the US Census Bureau’s 2015 American Community Survey (ACS). Share of aggregate income is from Table B19082 from the 2015 ACS. Share of aggregate energy consumption is calculated by grouping households according to income quintile’s upper limits, aggregating consumption in British Thermal Units (BTUs) for quintiles and the total, and calculating share of each group’s total to national total using the 2015 Residential Energy Consumption Survey (RECS). Share of aggregate residential energy emissions is calculated using different energy sources’ PM2.5 emissions intensity from Emissions & Generation Resource Integrated Database (eGRID), and each household’s consumption from the 2015 RECS. Rooftop solar panel deployment rate is calculated using the 2015 RECS data. Share of total residential energy credits is calculated using the 2015 Internal Revenue Service (IRS) Statistics of Income (SOI) data. County-level mortality damages per capita for 2017 using the AP4 model by Dennin et al. (2024) and is matched with county-level median income data from Table DP03 from the 2017 ACS 5-Year Estimates. Mortality damages are converted to 2020 dollars using the Gross Domestic Product (GDP) deflator from the Bureau of Economic Analysis’s (BEA) NIPA Table 1.1.9. Energy expenditure shares are calculated using the Public Use Micro Data (PUMD) files of 2015 Consumer Expenditure Survey (CES) data. Energy expenditure is calculated by summing up family level expenditures on electricity, natural gas, and other fuels used at home. The energy expenditure share is calculated by dividing energy expenditure by total expenditure for each family for each quarter and then averaging over quarters and for families that fall in each income quintile.

Table 2: Selected variables across the joint net worth and income quintiles from the data versus the model

NW-Inc Quintile	<u>Income</u>		<u>% Share of Expenditure</u>		<u>Wealth</u>		<u>% Ratio</u>			
	Data	Model	Data	Model	Data	Model	<u>Expend-Inc</u>		<u>Energy-Expend</u>	
	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model
Q1	9.9	9.6	13.0	13.5	-33.6	7.4	71.8	136.5	4.9	4.9
Q2	11.1	14.3	14.1	17.9	1.0	16.9	70.1	118.8	4.6	4.7
Q3	17.3	17.4	18.9	18.4	4.8	14.2	60.1	100.7	4.7	4.5
Q4	22.4	23.7	22.4	22.7	16.2	23.4	55.7	91.9	4.5	4.4
Q5	39.2	35.0	31.4	27.4	111.6	38.2	44.2	77.8	4.1	4.2

Notes: Data is from the 2010 PSID.